

L6a: Metabolic Engineering and Flux Balance Analysis

Stoichiometry, reaction bounds, and production limits
CHEME 5800 Cornell University

Objectives for Today

How do material balances and biological constraints limit the production of a desired compound?

Today's concepts

- You will **represent a metabolic reaction network** with a stoichiometric matrix and interpret its rows, columns, and coefficient signs.
- You will **formulate a flux balance analysis problem** by combining steady-state balances, reaction bounds, and a linear objective.
- You will **develop and interpret flux bounds** using reaction reversibility and enzyme capacity, and explain the assumptions that simplify the bounds model.

Lecture: [Metabolic Engineering and Flux Balance Analysis](#)

Lecture and Examples

Lecture: [Metabolic Engineering and Flux Balance Analysis](#)

In this lecture, we will represent a reaction network with a stoichiometric matrix and use material balances, flux bounds, and an objective to estimate the reaction rates.

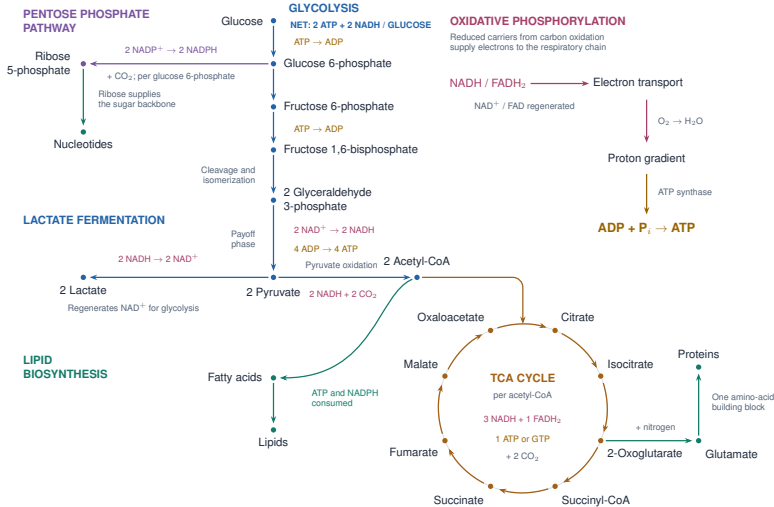
Example: [Calculate fluxes in a urea-cycle model](#)

In this example, we will estimate reaction directions and enzyme capacities, then solve a linear program to maximize urea export. We will use the material balances to identify the reaction capacity that limits production.

We can explore the connections among metabolic pathways in the [KEGG metabolic pathways map](#).

Central metabolism

Carbon flow, energy production, and biosynthetic branches



Selected pathways; arrows may combine reactions. Glycolysis and pyruvate-oxidation yields are per glucose; TCA yields are per acetyl-CoA. Cofactors shown are selective. Compartments, complete stoichiometry, and alternative routes are omitted; pathway availability depends on the organism.

What Is Metabolic Engineering?

Metabolic engineering is the directed modification of an organism's metabolic pathways through genetic and regulatory changes to increase production or enable the production of new products.

The engineering question is: how can we redirect carbon, nitrogen, and energy toward a target molecule while meeting the cell's other requirements?

A **metabolic flux** is a reaction rate expressed on a specified volume or biomass basis. Flux balance analysis estimates these rates subject to material balances and flux bounds, for a chosen objective.

Early work: [Bailey \(1991\)](#); [Stephanopoulos and Vallino \(1991\)](#).

Flux Balance Analysis

A metabolic reconstruction lists the reactions an organism can carry out. The list alone does not specify their rates.

1. We express metabolite balances as algebraic constraints.
2. We restrict reaction rates using flux bounds.
3. We choose an objective, such as maximizing product export, to select an optimal flux distribution.

Several flux distributions may satisfy the same constraints. The objective supplies the criterion for comparing them.

Metabolic Networks and Data Sources

The modeled network depends on the organism, the compartments included, and the purpose of the calculation.

- **Catabolic reactions** break down nutrients.
- **Anabolic reactions** synthesize cellular components and products.
- When we model transport between **compartments**, we represent a metabolite in the cytosol and the same compound in a mitochondrion as distinct species.

Reaction records establish stoichiometry. Kinetic and physiological data help specify the constraints for the conditions of interest.

Biological Data Resources

Resource	What we use it for
KEGG	We examine pathways and connections among genes, enzymes, and reactions.
BioCyc	We find pathway and genome information for a particular organism.
BiGG Models	We obtain metabolic models with standardized metabolite and reaction identifiers.
BRENDA	We find enzyme properties and kinetic measurements to estimate flux bounds.
BioNumbers	We find measured biological quantities and their literature sources.

Readings: Resource names link to the papers listed in the lecture.

Stoichiometric Matrix

Let $\mathcal{M}$ be the set of modeled species and $\mathcal{R}$ the set of reactions. The entry σ_{ij} is the net coefficient of species i in reaction j . We collect these coefficients in the matrix:

$$\mathbf{S} = [\sigma_{ij}] \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{R}|}.$$

- One **column** describes a reaction. One **row** collects contributions to a species balance.
- A positive σ_{ij} means reaction j produces species i ; a negative value means the reaction consumes it.
- A zero σ_{ij} means there is no net change in species i , including when its reactant and product coefficients cancel.

The coefficient signs refer to the written forward direction of each reaction. The matrix records net stoichiometry, not every catalytic or regulatory interaction.

One Reaction Gives One Matrix Column

Reaction j consumes one unit of A and two units of B to produce one unit of C . With rows ordered as (A, B, C) , the reaction column $\mathbf{s}_j$ is given by:

$$A + 2B \longrightarrow C, \quad \mathbf{s}_j = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}.$$

Let $\hat{v}_j$ be the reaction flux. Its contribution to the balances is:

$$\mathbf{s}_j \hat{v}_j = \begin{bmatrix} -\hat{v}_j \\ -2\hat{v}_j \\ \hat{v}_j \end{bmatrix}.$$

At positive flux, B is consumed at twice the reaction rate. For a reversible reaction, a negative flux reverses the production and consumption rates. The stoichiometric column stays the same.

The Flux Balance Approximation

We assume that intracellular metabolite amounts per unit biomass remain constant. We neglect explicit growth dilution and include transport and exchange reactions in the network.

The vector $\hat{\mathbf{v}}$ collects the reaction fluxes $\hat{v}_j$. For each balanced species $i \in \mathcal{M}$, production and consumption cancel across reactions $j \in \mathcal{R}$:

$$\sum_{j \in \mathcal{R}} \sigma_{ij} \hat{v}_j = 0, \quad \text{or} \quad \mathbf{S} \hat{\mathbf{v}} = \mathbf{0}.$$

Transport reactions contribute to the same balances as the other reactions.

Metabolite amounts per unit biomass remain constant even while reactions continue.

Derivation: [Balance assumptions and the conventional FBA approximation](#)

Flux Balance Analysis Formulation

Let c_j be the objective weight of reaction j , and let $\mathcal{L}_j$ and $\mathcal{U}_j$ be its prescribed flux bounds. With the objective weights and flux bounds fixed, the linear program is:

$$\begin{aligned} & \underset{\hat{\mathbf{v}}}{\text{maximize}} && \sum_{j \in \mathcal{R}} c_j \hat{v}_j \\ & \text{subject to} && \sum_{j \in \mathcal{R}} \sigma_{ij} \hat{v}_j = 0, \quad i \in \mathcal{M}, \\ & && \mathcal{L}_j \leq \hat{v}_j \leq \mathcal{U}_j, \quad j \in \mathcal{R}. \end{aligned}$$

A **feasible** distribution satisfies all balances and bounds. An **optimal** distribution attains the best objective value and need not be unique.

Readings: [Orth et al. \(2010\)](#); [Heirendt et al. \(2019\)](#).

An Objective for Product Export

Let p denote the urea exchange reaction, b4. We write this reaction as uptake from the surroundings, so **negative flux represents export**. We set its objective weight to -1 and all other weights to zero:

$$c_j = \begin{cases} -1, & j = p, \\ 0, & j \neq p, \end{cases} \quad \sum_{j \in \mathcal{R}} c_j \hat{v}_j = -\hat{v}_p.$$

Maximizing $-\hat{v}_p$ makes the urea exchange flux as negative as the balances and bounds allow, which maximizes urea export.

We express metabolic fluxes in $\text{mmol gDW}^{-1} \text{h}^{-1}$, where gDW means grams of dry biomass. Each bound has the units of its flux.

A Model for Flux Bounds

For enzyme-catalyzed reaction j , we combine the reference capacity $V_{max,j}^\circ$, relative enzyme abundance e/e° , activity factor θ_j , saturation factor f_j , and reversibility parameter δ_j to define the flux bounds:

$$\underbrace{-\delta_j \left[\overbrace{V_{max,j}^\circ \left(\frac{e}{e^\circ} \right) \theta_j(\dots) f_j(\dots)}^{\text{reverse: other functions or parameters?}} \right]}_{\mathcal{L}_j} \leq \hat{v}_j \leq \underbrace{V_{max,j}^\circ \left(\frac{e}{e^\circ} \right) \theta_j(\dots) f_j(\dots)}_{\mathcal{U}_j}.$$

This model assumes equal forward and reverse capacities for a reversible reaction. If the maximum rates differ, we need separate capacity estimates for the two directions.

We specify the parameters, enzyme abundances, and activity and saturation factors **before solving** the linear program.

Reference Enzyme Capacity

Let $k_{cat,j}^\circ$ be the reference catalytic turnover number and e° the characteristic enzyme abundance. The reference capacity is:

$$V_{max,j}^\circ = k_{cat,j}^\circ e^\circ.$$

Quantity	Meaning	Units
$V_{max,j}^\circ$	This is the maximum rate at reference enzyme abundance.	$\text{mmol gDW}^{-1} \text{h}^{-1}$
$k_{cat,j}^\circ$	This is the turnover rate at substrate saturation.	h^{-1}
e°, e	e° and e denote reference and actual abundance.	mmol gDW^{-1}

The ratio e/e° scales the reference capacity to the actual abundance of the enzyme catalyzing reaction j .

Activity, Saturation, and Reversibility

The activity factor θ_j , saturation factor f_j , and reversibility parameter δ_j are dimensionless:

Quantity	Meaning
$\theta_j(\dots) \in [0, 1]$	The activity factor gives the fraction of maximal activity available under allosteric regulation.
$f_j(\dots) \in [0, 1]$	The saturation factor describes substrate availability. A value of one represents saturating substrates.
$\delta_j \in \{0, 1\}$	The reversibility parameter is zero for a forward-only reaction and one when reverse flux is allowed.

Activity and substrate availability can reduce the capacity below $V_{max,j}^\circ(e/e^\circ)$.

Simplified Bounds Model

We assume the reference enzyme abundance, no allosteric effects, and saturating substrates. Setting $e/e^\circ = \theta_j = f_j = 1$ gives the bounds:

$$\underbrace{-\delta_j V_{max,j}^\circ}_{\mathcal{L}_j} \leq \hat{v}_j \leq \underbrace{V_{max,j}^\circ}_{\mathcal{U}_j}.$$

- For a **forward-only reaction**, $\delta_j = 0$, so the flux satisfies $0 \leq \hat{v}_j \leq V_{max,j}^\circ$.
- For a **reversible reaction**, $\delta_j = 1$, so the flux satisfies $-V_{max,j}^\circ \leq \hat{v}_j \leq V_{max,j}^\circ$.

We need estimates of the turnover number, characteristic enzyme abundance, and reaction reversibility.

Exchange Bounds Specify the Surroundings

Exchange bounds limit nutrient uptake and product secretion. In the urea example, we write every exchange as uptake from the surroundings, so **positive flux represents uptake** and negative flux represents export.

- The upper bound limits the uptake rate.
- An upper bound of zero prevents uptake through that reaction.
- A negative lower bound sets the largest allowed export rate.
- A lower bound of zero prevents export through that reaction.

The exchange bounds specify which nutrients the system can take up and which products it can release. Internal reactions have separate bounds based on enzyme capacity and reaction direction.

Example: Estimate Fluxes in the Urea Cycle

In this example, we will calculate the maximum urea export rate and identify the enzyme capacity that limits production.

1. We will build the stoichiometric matrix and estimate reaction directions from thermodynamic data.
2. We will calculate flux bounds from enzyme turnover numbers and an assumed enzyme abundance, and set the exchange bounds.
3. We will solve the linear program and use the material balances to explain which capacity limits urea export.

We will also examine how a competing reaction diverts material from urea production and changes the maximum export rate.

Example: [Calculate fluxes in a urea-cycle model](#)

Summary

We formulated flux balance analysis as a linear program, combining a metabolic network, biological constraints, and a production objective.

Key takeaways

- **Stoichiometric matrix.** We organized net production and consumption coefficients into a matrix to write the material balances.
- **Flux balance analysis.** We combined material balances, flux bounds, and a linear objective to estimate an optimal flux distribution, which need not be unique.
- **Flux bounds.** We used enzyme capacity and reaction direction to limit the allowed rates. We simplified the bounds by assuming reference enzyme abundance, no allosteric effects, and saturating substrates.

The same reaction network can support different production rates under different constraints. An optimal solution describes what the model allows under the conditions we specify.